ELE 420/520 — Project 4: Zeek MITM Detection

This PDF includes **detect_mitm.zeek**, **no_mitm.log** (offline pcap, no attack), and **mitm.log** (offline analysis of Project 3 MITM capture mitm.pcapng showing ALERT lines). Replace **LastName** and **FirstName** in the filename before submitting. Add your Mininet console snapshot as an additional page or embed it in your final PDF if required.

1. detect_mitm.zeek

```
global DATA_AGG_IP: addr = 10.0.0.20;

# dictionary-like data structure to store measurements received by relays
global read_from_relays : table[count] of double = {
    [22] = 0.00, [32] = 0.00, [42] = 0.00, [52] = 0.00, [62] = 0.00, [72] = 0.00, [82] = 0.00, [92] = 0.00,
    [13] = 0.00, [23] = 0.00, [33] = 0.00, [43] = 0.00, [53] = 0.00, [63] = 0.00, [73] = 0.00, [83] = 0.00, [93] = 0.00,
};
global read_from_relays_count : count = 0;

# dictionary-like data structure to store measurements sent to control center
global read_from_da : table[count] of double = {
    [22] = 0.00, [32] = 0.00, [42] = 0.00, [52] = 0.00, [62] = 0.00, [72] = 0.00, [82] = 0.00, [92] = 0.00,
    [13] = 0.00, [23] = 0.00, [33] = 0.00, [43] = 0.00, [53] = 0.00, [63] = 0.00, [73] = 0.00, [83] = 0.00, [93] = 0.00,
};
global read_from_da_count : count = 0;

# By making this variable to T, you can let zeek to print some measurements from different event handlers
global DEBUG : bool = F;

# Note that here we don't use the Zeek's logging framework; just a simpley log file for our experiment
global dnp3m_log = open("dnp3m.log");

function table_equal(t1: table[count] of double, t2: table[count] of double) : bool
    {
    for (i, m in t1)
        if ( !(i in t2) )
            return F;
        else
            if ( t2[i] != m)
                return F;
    return T;
    }

event zeek_init()
    {
    if (DEBUG)
        print |read_from_da|;
    }

event zeek_done()
    {
    print dnp3m_log, read_from_relays;
    print dnp3m_log, read_from_da;
    close(dnp3m_log);
    }
event dnp3m::request(c: connection, is_orig: bool, index: vector of count)
    {
    if (DEBUG)
        print "dnp3m request", c$id, is_orig, index;
    }

event dnp3m::response(c: connection, is_orig: bool)
    {
    if (DEBUG)
        print "dnp3m response", c$id, is_orig;
    }

event dnp3m::data(c: connection, is_orig: bool, index: count, measure: double)
    {
    if (DEBUG)
        print "dnp3m response data", c$id, is_orig, index, measure;

    if (c$id$orig_h == DATA_AGG_IP)
        {
        if (index in read_from_relays)
            {
            # this if branch corresponds to packets from relays; store "measure" in the appropriate table with the "index"
            read_from_relays[index] = measure;

            read_from_relays_count = read_from_relays_count + 1;
            }
        }

    if (c$id$resp_h == DATA_AGG_IP)
        {
        if (index in read_from_da)
            {
            # this if branch corresponds to packets sent to control center; store "measure" in the appropriate table with the "index"
            read_from_da[index] = measure;

            read_from_da_count = read_from_da_count + 1;
            }
        }

    # I use a very naive way to determine when to compare measurements just for this experiment
```

```
#  when zeek observs that the data_aggregator sends all the measurements, it will compare the measurements
if (read_from_da_count == |read_from_da|)
    {
    if (read_from_da_count != read_from_relays_count)
        print dnp3m_log, "ERROR on parsing";

    if( !table_equal(read_from_relays, read_from_da))
        print dnp3m_log, "ALERT: data collected from relays are not consistent with the data sent to the control center";

    read_from_da_count = 0;
    read_from_relays_count = 0;
    }
}
```

2. no_mitm.log (zeek on data_aggregator.pcap)


```
{
[62] = 1.93,
[73] = -100.0,
[33] = 85.0,
[23] = 163.0,
[83] = 0.0,
[32] = 4.77,
[42] = -2.41,
[22] = 9.67,
[53] = -90.0,
[72] = 0.62,
[52] = -4.02,
[43] = 0.0,
[82] = 3.8,
[93] = 0.0,
[13] = 71.949997,
[92] = -4.35,
[63] = 0.0
}
{
[62] = 1.93,
[73] = -100.0,
[33] = 85.0,
[23] = 163.0,
[83] = 0.0,
[32] = 4.77,
[42] = -2.41,
[22] = 9.67,
[53] = -90.0,
[72] = 0.62,
[52] = -4.02,
[43] = 0.0,
[82] = 3.8,
[93] = 0.0,
[13] = 71.949997,
[92] = -4.35,
[63] = 0.0
}
```

3. mitm.log (zeek on project3/mitm.pcapng)

```
ALERT: data collected from relays are not consistent with the data sent to the control center
{
[32] = 0.09,
[92] = -0.07,
[93] = -1.25,
[13] = 0.67,
[72] = 0.01,
[53] = -0.9,
[22] = 0.17,
[62] = 0.04,
[82] = 0.07,
[83] = 0.0,
[23] = 1.63,
[42] = -0.04,
[43] = 0.0,
[52] = -0.07,
[63] = 0.0,
[33] = 0.85,
[73] = -1.0
}
{
[32] = 0.1,
[92] = -0.08,
[93] = -1.49,
[13] = 0.67,
[72] = 0.01,
[53] = -1.07,
[22] = 0.19,
[62] = 0.04,
[82] = 0.08,
[83] = 0.1,
[23] = 1.79,
[42] = -0.04,
[43] = 0.23,
[52] = -0.08,
[63] = -0.11,
[33] = 1.02,
[73] = -1.14
}
```